Udyog Saarthi: A Progressive Web Application for Accessible Job Guidance for Adults with Disabilities

Annie Mary Joseph
Dept. of CSE
Presidency University
Bengaluru, India

Jozach Rajan Samual
Dept. of CSE
Presidency University
Bengaluru, India

Khushi Yadav
Dept. of CSE
Presidency University
Bengaluru, India

Dr. Sridevi S
Assistant Professor, Dept. of CSE
Presidency University
Bengaluru, India

***Abstract*—Adults with Intellectual Disabilities (ID) continue to be left out of meaningful work opportunities in India, even though reservation quotas legally exist. The problem is not that jobs are unavailable, but that the way job information is currently published makes it extremely hard to understand and act on. Government portals contain the right information, but are not built for users who find long text, complex forms, and technical jargon difficult to navigate.**

Udyog Saarthi is our attempt to fix that gap. It is a Progressive Web App (PWA) that pulls government job notifications into a single place, and converts them into plain, short, simple language that adults with ID and their caregivers can actually understand. We built this system, tested multiple interaction pathways, and removed features that were too heavy or impractical on typical low-end phones (such as face recognition and on-device Sign Language processing). Early testing with a small group of users and caregivers showed better comprehension of job requirements and easier navigation across available opportunities. This suggests that simplifying job information can directly improve job discovery and participation for this ignored population.

***Index Terms*—intellectual disability, assistive tech, accessibility, progressive web app, simplified language, job discovery**

I. Introduction

Getting a job is about much more than income; it is about independence, routine, and social participation. In India, adults with Intellectual Disabilities (ID) are entitled to reservation benefits under the Rights of Persons with Disabilities (RPwD) Act [2], yet many still struggle to convert those legal rights into real opportunities. The barrier is often not the lack of vacancies but the way information is presented: job notices are long, dense, and scattered across portals. They assume high reading proficiency, confident web navigation, and familiarity with government forms—assumptions that exclude precisely the people the policy aims to include.

This paper addresses a simple question with practical consequences: if we bring public-sector vacancies into one place and present them in plain, step-by-step language with accessibility-first UI, do adults with ID discover and understand relevant jobs more effectively? We designed and built Udyog Saarthi, a Progressive Web App (PWA) that aggregates reservation-category vacancies published on official sources (e.g., NCS) and delivers them through simplified summaries, larger touch targets, high-contrast modes, and caregiver-mediated actions [6]. The goal is not to "teach users to use complex portals" but to reduce the cognitive load of the portal itself.

During prototyping, we explored several advanced interaction ideas—such as face-recognition login and Indian Sign Language support—but found that their computational and operational overheads on common low-end Android phones outweighed their benefits in real-world conditions. We therefore dropped these directions and focused on what consistently helped users: cleaner information architecture, plainer language, and predictable task flows [7]. We report these negative results because they shaped our final design and may save others time and resources.

Our early evaluations with trainees and caregivers emphasize an important design lesson: when job information is consolidated and written for the reader—not the issuing department—people complete core tasks faster and with fewer errors. This is a design problem before it is a training problem.

Contributions. This paper offers:

1) a needs-driven analysis of job-discovery barriers faced by adults with ID in India's public-sector hiring context;
2) a PWA architecture that consolidates reservation-category vacancies and delivers accessibility-first, plain-language summaries;
3) a practical account of discarded alternatives (face recognition, ISL processing) and why lightweight designs worked better on low-end devices; and
4) preliminary evidence from user tasks indicating improved comprehension of eligibility and application steps when using Udyog Saarthi.

II. Related Work

Several digital infrastructures already exist in India to support job discovery, but they are not designed for people who struggle with dense text, complex navigation, or abstract forms [5]. The National Career Service (NCS) portal is the primary official channel where government vacancies are published, including reservation-category postings. However, its structure assumes that users can read multi–page PDFs, interpret formal eligibility wording, and follow nested menu hierarchies. This makes the experience cognitively heavy for adults with Intellectual Disabilities (ID), especially for those with limited literacy.

There are also non-profit and private initiatives around disability employment support, but they tend to focus on industry exposure, training programs, or private-sector openings. They do not centralize public-sector posts, and importantly, they do not simplify the job text itself. In almost all cases, the user is expected to adapt to the interface, rather than the interface adapting to the user.

Our work is positioned differently: instead of training users to handle complexity, we reduce the complexity of the interface and the information. This "cognitive load first" framing has been underexplored in India in the context of job discovery for adults with ID. Table I summarizes the gap in existing systems.

TABLE I
COMPARISON WITH EXISTING JOB-ACCESS PLATFORMS

Platform / Service	Govt Jobs	Unified View	Plain Language
NCS (India)	Yes	Partial	No
NGO Initiatives	Limited	No	Partial
Udyog Saarthi (ours)	Yes	Yes	Yes

This narrow-scope review is intentional. Our problem space is extremely context-specific to Indian public-sector hiring [3]. Most prior efforts look at employment training or placement; very few examine the job information design itself as the barrier. Our work takes this gap as the starting point for system design.

III. PROBLEM DEFINITION & REQUIREMENTS

Adults with Intellectual Disabilities (ID) who are preparing for employment face two simultaneous barriers in the digital job search process: discovering where relevant government vacancies are posted, and understanding what those postings are actually saying. Both of these steps must succeed for a job opportunity to become actionable. In practice, these steps often fail at the very first page load—because the job information is scattered across multiple government sources, and written in language that assumes high reading proficiency.

We therefore define the core problem addressed by Udyog Saarthi as: enabling adults with ID to both find reservation-category government jobs and understand the eligibility and steps required to apply.

From formative interviews, observation sessions, and feedback from caregivers and trainers, the following design requirements were derived:

- consolidate government vacancies into a single interface so that the user does not need to visit or understand multiple portals
- present information in short, plain sentences, with step-wise break-down of eligibility conditions and required documents
- keep the UI visually and cognitively simple: minimal choices, predictable patterns, large touch targets, and high-contrast options [4]
- support caregiver participation, because many users rely on guided actions or joint decision-making
- ensure fast performance on low-end Android devices, because that is the dominant hardware profile for this population

In short, the system must reduce the amount of work required from the user, not increase it. The objective is not to "train" users into expert navigators of heavyweight government portals, but to redesign how information is delivered—so that jobs that already exist become actually reachable.

IV. SYSTEM DESIGN & ARCHITECTURE

Udyog Saarthi is designed around one principle: the system should do the complex work so the user does not have to. The architecture is therefore intentionally lightweight on the client side, and more active on the backend. At a high level, it has three major layers: (1) data ingestion and normalization, (2) backend processing and storage, and (3) the PWA client that users interact with.

A. Data Ingestion and Normalization

Official government portals and notices do not have uniform structures. Some use PDFs, some have static HTML, some require navigation through multiple click paths. We use scheduled scraping pipelines to fetch reservation-category job posts from official sources (primarily NCS) at regular intervals. These posts are parsed, cleaned, and converted into structured fields (dates, eligibility criteria, category, state/department, links to original notification). This reduces ambiguity and ensures all content entering the system has a predictable shape.

B. Backend Processing

Once the scraped data is normalized, the backend generates two parallel representations of each job posting: (1) the original text, preserved for fidelity, and (2) a simplified plain-language summary. We use controlled language transformations—short sentences, bullet points, reduced jargon—to lower cognitive load. This backend also manages caregiver-linked user accounts, stores job history, and triggers notification updates when new vacancies appear.

C. PWA Client

The frontend is built as a Progressive Web App to ensure installability, offline caching of last-seen posts, and fast loading even on low bandwidth. The interface prioritizes clarity over feature density. Users can browse simplified summaries, enlarge text, switch to high-contrast mode, and ask caregivers to assist in the final steps. Instead of offering every option at once, the UI presents one step at a time and keeps interaction flows shallow (no deep menu hierarchies).

In combination, this architecture allows the system to handle complexity behind the scenes, and present a clean, low-effort, low-reading-demand experience to adults with ID and their caregivers.

V. Exploration of Alternatives

During early prototyping, we explored several advanced interaction ideas that appeared promising on paper but failed under realistic constraints of the target population and their hardware. We include these not as features, but as negative results that shaped the final design.

A. Face Recognition Login

We considered passwordless login using face recognition to reduce the need for typing and remembering credentials. While technically feasible, the approach had two major problems: (1) on-device processing on low-end Android phones introduced visible lag and inconsistency, and (2) caregiver-assisted account management was already smoother and more familiar to users than biometric enrolment. After internal tests, we concluded that face recognition added friction instead of removing it.

B. Indian Sign Language (ISL) Integration

We experimented with early ISL translation concepts, including the idea of avatar-based sign language output. However, the compute and memory requirements were not compatible with our target devices and deployment model. Approaches based on server-side video assets also created latency and bandwidth issues. In short: the implementation cost was high, and the practical benefit for this specific user group (whose primary barrier was cognitive load, not deafness) was not as high as simplifying text itself.

These discarded directions taught us the most important lesson of the project: fancy features do not automatically make software more accessible. The best accessibility gain came not from heavy computer vision or media generation, but from lowering reading and navigation complexity.

VI. Implementation Details

Udyog Saarthi was implemented using a modern web stack chosen for reliability, fast iteration, and ability to run smoothly on low-end hardware.

A. Frontend

The client is a Progressive Web App built using React. We intentionally kept the visual styling minimal, used large tap targets, and provided quick toggles for text sizing and contrast. Components are structured so that each core user action (reading eligibility, saving a job, sharing with a caregiver) appears as a single clear step rather than multiple nested interactions. Service workers are used to cache the most recent vacancy list and static assets, enabling functional use even during poor connectivity.

B. Backend

The backend uses Node.js for API orchestration and PostgreSQL for persistence. We maintain two versions of each job posting in the database: the original text (for traceability and auditing) and the simplified summary. Cron schedules trigger Python-based scrapers periodically to fetch and parse new postings. The normalization scripts extract structured fields (e.g., application deadlines, category tags, document requirements) so the UI does not have to deal with raw text processing.

C. Caregiver Mediation Layer

User accounts are designed to allow caregivers to assist with decisions. A caregiver can link to multiple beneficiaries, monitor saved jobs, or help finalize application steps. This reflects the real-world support structure of many ID households and avoids isolating the user.

This implementation strategy allowed us to maintain technical feasibility while still centering accessibility and low cognitive load.

VII. Evaluation

We conducted a small formative evaluation with ten users: a mix of adults with Intellectual Disabilities and their caregivers. The purpose was not to measure placement outcomes, but to understand whether the redesigned interaction model actually reduced effort in the two foundational steps: finding relevant vacancies and understanding what they required.

Participants were given simple tasks such as:

- identify at least one public-sector job that they were eligible for, and
- describe in their own words what documents or criteria were needed to apply.

Across the group, users were noticeably faster and more confident when interacting with simplified summaries versus original text. Several users who initially hesitated to read the entire job description were able to complete the task once the information was broken down into short, plain sentences. Caregivers also reported lower time cost per user when assisting through the PWA, in comparison to navigating government websites directly.

The takeaway from this early study is practical and modest: the accessibility-first restructuring of job information shows clear signs of improving comprehension and reducing cognitive load. While this is not yet a large-scale evaluation, the pattern is consistent and supports the hypothesis that simplifying information design can improve job discovery for this population.

VIII. Discussion

One of the clearest insights from this work is that accessibility for adults with Intellectual Disabilities is not primarily a matter of adding assistive "features"—it is about removing unnecessary complexity from the main experience. When we stopped focusing on advanced modalities (like biometrics or sign-language avatars) and instead concentrated on reducing reading and navigation load, users were more successful with far less training.

This also reframes disability inclusion for government job portals in India. The dominant assumption in mainstream systems is that the user should adjust to the platform. The more appropriate model for this population is the reverse: the platform must adjust to the cognitive abilities and working memory limits of the user. Caregiver-mediated workflows reinforce this point: the support system around the user is a legitimate part of the interaction model, not a workaround.

The broader suggestion for policy and technology design is that inclusion in employment requires inclusion in information access first [1]. If reservation-policy benefits are locked behind dense PDFs, they effectively disappear for the people they are intended to reach. Designs like Udyog Saarthi demonstrate that small but intentional language and interaction shifts can turn formal access into practical access.

IX. Future Work

There are several directions that can extend this work. First, we plan to increase the geographic and institutional coverage of vacancy sources by integrating more state-level government feeds and by pursuing formal API partnerships with agencies where possible. This will reduce reliance on scraping and improve data freshness.

Second, we intend to expand personalization. Different users benefit from different forms of simplification, and a one-style-for-all approach may not be optimal. An adaptive summarization layer—one that can vary the length or specificity of the explanation depending on the user's reading comfort—could make the system more effective.

Third, we want to incorporate voice pathways, both for navigation and input. Many users are more comfortable listening and speaking than reading and typing. Lightweight speech-to-text and text-to-speech flows could reduce barriers further, if implemented in a local, low-bandwidth compatible form.

Finally, a larger field trial across multiple vocational training environments can provide stronger evidence on whether these improvements in comprehension actually influence real job applications over time. Impact measurement is essential: inclusion technology must not only be usable, but measurably useful.

X. Conclusion

Udyog Saarthi addresses a very specific but highly consequential gap: adults with Intellectual Disabilities in India have legal access to reservation–category public-sector jobs, but not practical access to the information they need to apply for them. By consolidating vacancies into one place and translating them into plain, short, stepwise language, the system reduces the cognitive effort needed to discover and understand job opportunities.

Our findings suggest that accessibility for this population is most effectively improved by simplifying information design, not by adding complex, hardware-intensive features. Negative results from abandoned ideas—such as face recognition and Indian Sign Language processing—reinforce this principle.

This work demonstrates that employment inclusion begins upstream: with how job information is written, structured, and delivered. Even modest interface changes can make real opportunities visible and reachable to the people the policies were created for. Udyog Saarthi is one practical step in that direction.

Acknowledgment

We thank NIEPMD and the accessibility experts, caregivers, and trainees who contributed to design workshops and usability testing.

References

[1] World Health Organization. World Report on Disability. WHO Press, 2011.

[2] Government of India, Ministry of Law and Justice. The Rights of Persons with Disabilities Act (RPwD), 2016.

[3] Department of Personnel and Training, Government of India. Reservation for Persons with Benchmark Disabilities in Posts/Services under Central Government. DOPT, Office Memorandum, 2018.

[4] W3C. Web Content Accessibility Guidelines (WCAG) 2.1. W3C Recommendation, June 2018.

[5] A. A. Parnandi and G. Floyd. "Assistive User Interface Design for People with Intellectual Disabilities: A Review." ACM SIGACCESS Conference on Computers and Accessibility (ASSETS), 2021.

[6] A. Russell and J. Song. "Progressive Web Apps: Bridging Mobile Experiences across Environments." Communications of the ACM, vol. 62, no. 4, pp. 22–24, 2019.

[7] B. Plimmer and D. R. Inglis. "Plain Language for People with Cognitive Impairments: Rethinking Text Design." Proceedings of the ACM CHI Conference on Human Factors in Computing Systems, 2020.